



Burkina Institute of Technology

BURKINA FASO

La Patrie ou la Mort, nous
Vaincrons

KOUDOUGOU

Workshop

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Course: Business Application/ AFRICA CASE

Topic: Frequent Electricity Outages

Problem description

In Burkina Faso, and particularly in Ouagadougou, power outages are a recurring phenomenon, especially during the hot season when demand often exceeds the production capacity of SONABEL (the national electricity company). These outages occur frequently without reliable warning, disrupting the daily lives of households, the operation of small businesses, and the activities of students who rely on electricity to study in the evening. Outages can last anywhere from a few minutes to several hours, and their unpredictability makes it difficult for people to organize effectively. Shopkeepers lose perishable goods (cold drinks, frozen products), students lose valuable study time, and families often have to scramble to buy candles or fuel in an emergency. This problem affects a large portion of the urban and peri-urban population, with a real economic and social impact. The lack of reliable information on outage schedules makes the situation worse, since people can neither anticipate nor adapt. This context creates a clear need for a simple, accessible tool to help manage this uncertainty on a daily basis.

Target Users

- Urban and peri-urban households dependent on the electrical grid
- Small businesses (shops, restaurants, vendors of fresh/frozen products)
- Students and pupils studying in the evening
- Small enterprises and workshops using electrical equipment

Current Alternatives

- Use of power generators (high fuel cost, noise, pollution)
- Solar lamps or flashlights, often of inconsistent quality
- Candles, which are unsafe and costly in the long run
- Word-of-mouth or informal social media groups to find out which areas are affected
- No real anticipation possible due to the lack of centralized information

Proposed Solution (Digital) A mobile/web application — also accessible via SMS for areas with weak connectivity — that centralizes information on power outages. It would combine:

- Official announcements of planned outages (if available via SONABEL)
- A real-time community reporting system (users report an outage in their neighborhood)
- Zone/neighborhood-based notifications to warn users
- A history of outages by neighborhood to help people anticipate them (e.g., trends by time of day/day of week)

Business Potential

- **Freemium model:** free version with basic alerts, premium version with personalized neighborhood notifications, advanced statistics, and early alerts
- **Partnerships:** selling aggregated (anonymized) data to SONABEL or NGOs to improve network planning
- **Targeted advertising:** partnerships with sellers of solar panels, batteries, or generators (a fast-growing local market)
- **Mobile Money payment:** premium subscription payable via Orange Money/Moov Money

